Almanac 47 v2.3.0

Celestial navigation on the SwissMicros C47 / R47 — user manual — version 2.3.0 (Local Time)

What is new in v2.3.0 "Local Time"

- **UT or local time on +/-.** Press +/- on the menu or on any view: every time shown switches between UT and local time, LT = UT + TZ. Store your zone once in the variable **TZ**, in hours east: 3 STO "TZ" for UTC+3, -5 STO "TZ" for UTC−5 (section 4).
- In LT the header shows the local date and time and LT; rise, set, twilights and meridian passage are in LT. GHA, Dec, Hc and Zn do not change, and DATE and UTC are still asked in UT. The choice stays for the next run (the variable TZLT).
- **A new header on every view:** the date and time on the left, the view's name in the middle, the position on the right.
- **MOON47:** +/- shows its times in UT or LT; the view from the south moved to the key **0**.
- Everything else as v2.2.1: the same results and the same pages.

Almanac 47 puts an almanac page on the C47: GHA, declination, Hc and Zn of the Sun, the Moon, the planets and the navigation stars for your DR and time, sunrise, sunset, twilight and the Moon's phase, with horizon charts of the sky. Everything is computed on the calculator (VSOP87, Meeus, IAU precession and nutation).

This manual is for the C47 and the R47 (and the DM42n with the C47 firmware). ALMA47 is written with the C47's structured programming commands — IF … ELSE … ENDIF, DO … WHILE … ENDDO, REPEAT … UNTIL — and has no GTO at all; Free42, Plus42, the DM42 and HP-42S firmware do not have these commands. It carries all the speed work done since v2.1.0: the same pages, drawn with far fewer steps (section 6). **Free42 / Plus42 / DM42 with the stock firmware:** the zip Almanac47_Free42_DM42_v2.3.0.zip (the programs of v2.1.0, keeping your stack, with UT / LT on +/-) and its manual, Almanac47_Free42_Manual.pdf.

New in v2.2.1: ALMA47 and MOON47 give you back your stack (X … D of the 8 levels) and LASTx when they end, and use only part of the numbered registers (ALMA47 R07–R98, MOON47 R07–R65; the others stay yours). On ALMANAC and SPLIT, a key 1–8 picks that row of the table and puts the body on your stack: Z its text, Y Zn, X Hc (section 4). The stack is saved with **LocR 9**, **STOS R.00** and given back with **RCLS R.00**, a trick from the SwissMicros forum (section 4). The pages are those of v2.2.0, pixel for pixel.

It supports, and does not replace, the Nautical Almanac. The menu and INFO say so.

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Part 1 — using ALMA47

1. What to load

File	What it is
ALMA47	the programs that stay on the calculator: the menu and every view. On the calculator only ALMA47 has a name; its routines are N01, N02 … (section 7)
NAVINIT_FAST	the program NAVINIT: builds the matrices of a series fitted to 2026–2030 (smaller and faster)
NAVINIT_FULL	the program NAVINIT: builds the matrices of the full series, valid 2000–2050 (VSOP87 and Meeus, about twice the memory). Load NAVINIT_FAST or NAVINIT_FULL
MOON47	the Moon phase on its own, from the clock (section 8); needs nothing else
ALMA47_LABELS.txt	the labels of ALMA47 and the routines you can call yourself

File	What it is
p47u/	the same programs as rejig sources (.p47u) with comments, for the PROGRAMS folder of the C47

Each comes as .p47 (load it as it is) and as .txt (the listing). The old DM42 with the C47 firmware (64 KiB) has its own zip with NAVLITTLE (section 9). The almanac tables (TBL) of earlier versions are not part of v2.2 / v2.3 (they stay in the repository).

2. First start

- The C47 needs a firmware with **ATEXT**, **GRFNT** and the **STRUCT** commands: 00.109.05.00a0.ALPHA (5 Oct 2026) or later. The screens write their text with ATEXT in GRFNT 21 (the tinyFont, GRFNT 10, on the charts); only the body symbols are drawn with AGRAPH. On the older firmware 00.109.04.00b0 use the v1.1.0 release.
- Load ALMA47.p47 and one NAVINIT file (USB disk, then READP in the I/O menu). The .p47u files convert with **rejig -o ALMA47.p47 ALMA47.p47u** (a rejig that knows the C47 STRUCT items writes their numbers, IF 01 …); their comments come after a ";" and rejig drops them, so they cost no memory.
- **XEQ "NAVINIT"** once: it builds the matrices and shows MATRICES READY: FAST 2026-2030 (or FULL 2000-2050). Then delete it (GTO "NAVINIT", DELP): the matrices stay. Do not use CLPALL: it deletes ALMA47 too.
- ALMA47 asks DATE in the calculator's date format (CLK menu: YYYY-MM-DD, DD.MM.YYYY or MM/DD/YYYY) and reads it with the C47's own date functions; the prompt shows which.
- What changed inside is in OPTIMIZATIONS.md (in the zip, docs/ in the repository).

3. Starting ALMA47

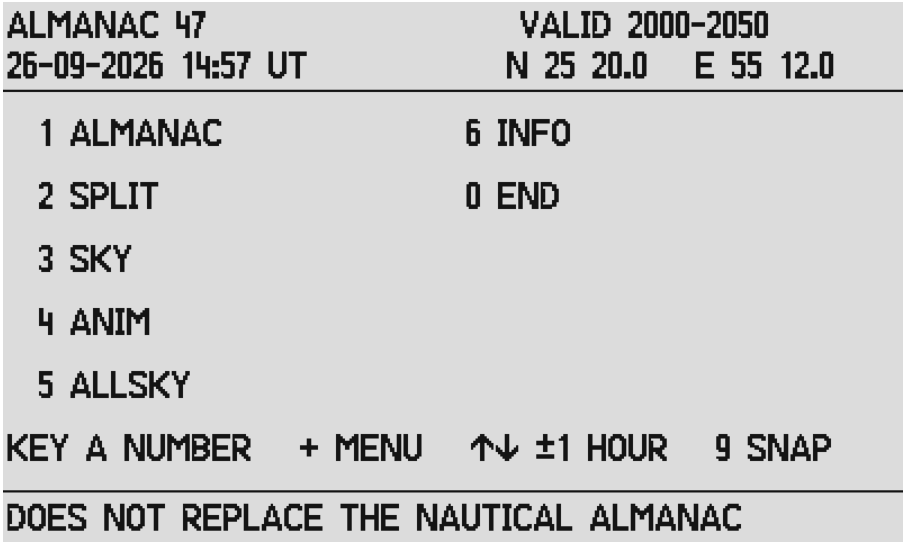
XEQ "ALMA47" asks four numbers, once, with INPUT (R/S keeps the value shown). Your stack is saved first; at each prompt the stack is clear and Y shows only the format of that input: DATE YYYY.MMDD (DATE DD.MMYYYY or DATE MM.DDYYYY when the CLK date format is D.MY or M.DY), UT HH.MMSS, LAT DD.MMm S -, LON DDD.MMm W -.

Input	Meaning	Example
DATE	UT date in the CLK date format: YYYY.MMDD, DD.MMYYYY or MM.DDYYYY	2026.0926
UTC	UT, HH.MMSS (UT1; add DUT1 to UTC for full accuracy)	14.57
LAT	latitude DD.MMm, south negative	25.20 = 25° 20′ N
LON	longitude DDD.MMm, west negative	55.12 = 55° 12′ E

After LON the sky is computed (the Sun, the Moon, the planets, the stars, the sun times) and kept in the matrix ALMC; then the menu appears, and the views only draw. While the sky or a view is computed, the hourglass shows in the top right corner and a box in the middle of the screen says SINKING....ABOUT; the view appears when it is complete. ALMC stays between runs: with the same date, time and place the menu and the views come at once.



4. The menu

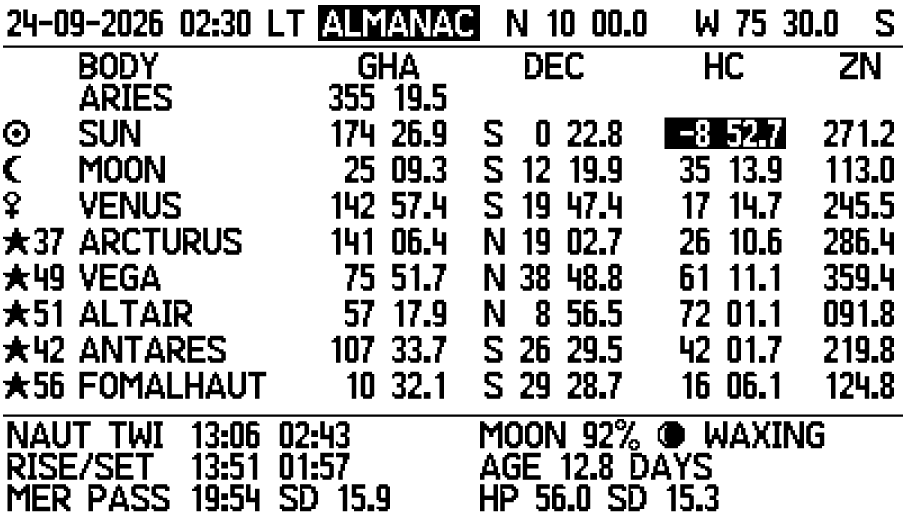


Title ALMANAC 47 and the period of the matrices; under it the date, the time (UT, or LT after +/-) and the position in use. Under the items, the keys: + MENU, ↑↓ ±1 HOUR and 9 SNAP; +/- is not written there.

Key	On the menu	On a view
1 – 6	the item is highlighted and its view is drawn (6: INFO)	—
9	SNAP (a screenshot)	SNAP (a screenshot)
+	—	back to the menu
▲ / ▼	one hour later / earlier (the title shows the new time)	the same view one hour later / earlier
1 – 8	—	ALMANAC and SPLIT: ALMA47 ends with that row on your stack (Z the text, Y Zn, X Hc)
+/-	UT or local time (LT = UT + TZ): the menu is drawn again	the same: the view is drawn again in UT or LT
0	end of ALMA47: your stack and LASTx back, R07–R98 cleared, the normal screen	—
other keys	ignored	ignored (R/S or EXIT stop the program)

The hours added with the arrows stay for the other views. New date or position: XEQ "ALMA47" again.

UT or local time (+/-). Store your time zone once in the variable **TZ**, in hours east of Greenwich: **3 STO "TZ"** for UTC+3, **-5 STO "TZ"** for UTC−5, 5.5 for UTC+5:30 (MOON47 uses the same TZ). Then +/- on the menu or a view switches every time shown between UT and LT = UT + TZ: the header shows the local date and time and LT, and rise, set, twilights and meridian passage are given in LT (the events of the UT day). The choice is the variable **TZLT** (0 UT, 1 LT) and stays for the next run. DATE and UTC are still asked in UT, and GHA, Dec, Hc and Zn do not change. Without TZ, ALMA47 creates TZ and TZLT with 0.



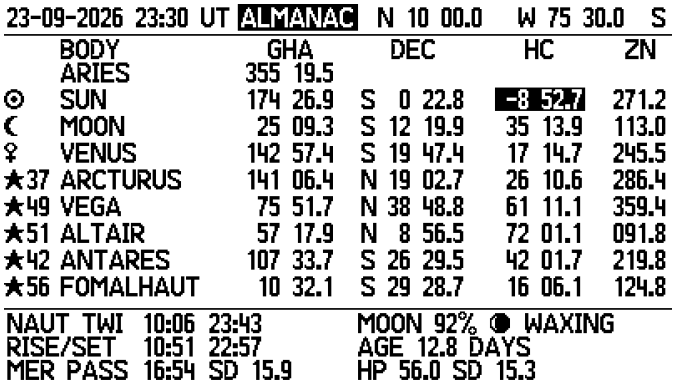
ALMANAC in LT with TZ 3: the same moment as section 5 (23:30 UT) shown as 02:30 LT on 24 Sep; the sun times +3 h.

A body on your stack (keys 1–8). On ALMANAC or SPLIT, press the number of a row (1 is the first body of the table, the Sun): ALMA47 ends and pushes three values on your stack — Z a text such as "SUN Zn 216.7 Hc 54 11.6", Y the azimuth Zn and X the altitude Hc (decimal degrees) — ready for a sight reduction. Your own X, Y … are above them. A row that is not there does nothing.

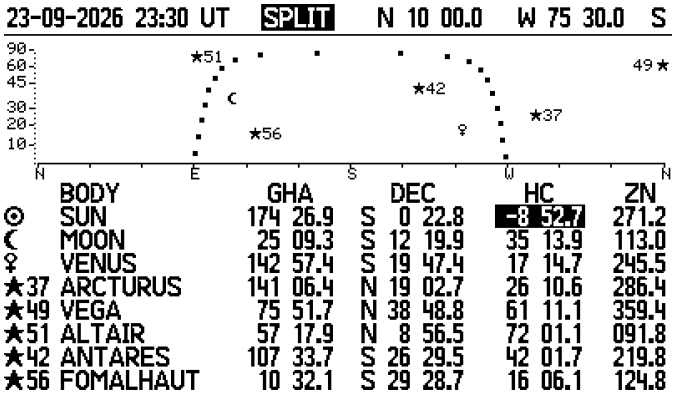
Your stack and registers. ALMA47 saves your stack (the 8 levels, X … D) and LASTx when it starts and gives them back when it ends, with 0 or with a key 1–8. It does it in one step each way, the trick shared on the SwissMicros forum: **LocR 9** (nine local registers, R.00–R.08), **STOS R.00** (the stack into R.00–R.07), LASTx into R.08; at the end RCL R.08, STO L, **RCLS R.00** — and the local registers go away with the program's return. It uses the numbered registers **R07–R98** while it runs and clears only those at the end (R-CLR 7.92): **R00–R06 and R99 stay yours.** It also puts your stack size back and the standard font (GRFNT 20). Named variables and matrices are not touched (DATE, UTC, LAT, LON and the matrices of NAVINIT stay). If you stop ALMA47 with R/S or EXIT, the registers keep its values (that is how the routines of section 7 can be called from the keyboard) and the stack is ALMA47's.

5. The views

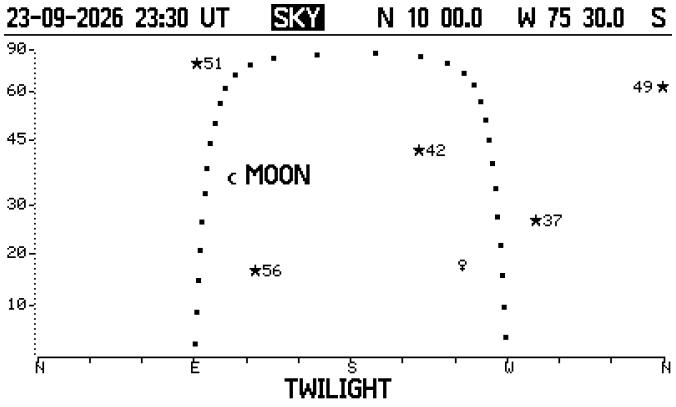
All views: 23 Sep 2026 23:30 UT, 10° N 075° 30′ W (the menu: 26 Sep 2026 14:57 UT, 25° 20′ N 055° 12′ E). Every view starts with the same line: the date and time (UT or LT), the view's name, the position, and T / S (tables / series) at the right.



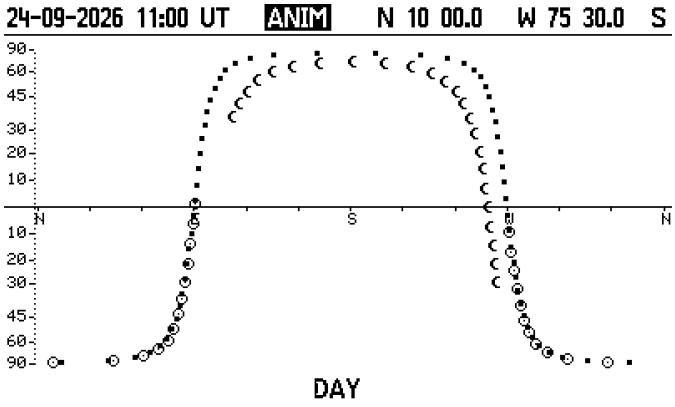
1 ALMANAC (ALMF): GHA ARIES, then GHA, Dec, Hc, Zn of 8 bodies — the Sun, the Moon if it is up, the first planet above the horizon, the brightest stars higher than 10°; twilight, rise/set, meridian passage, the Moon (lit part, phase symbol, age, HP, SD).



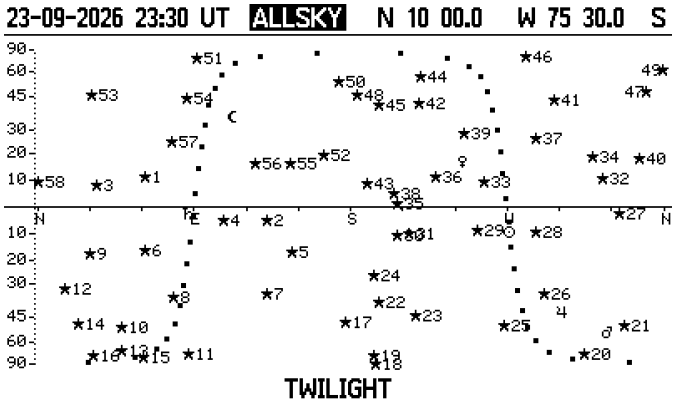
2 SPLIT (HALMH): the chart on top (Hc up, Zn across, the celestial equator dotted), the bodies below down to the bottom of the screen (8 rows).



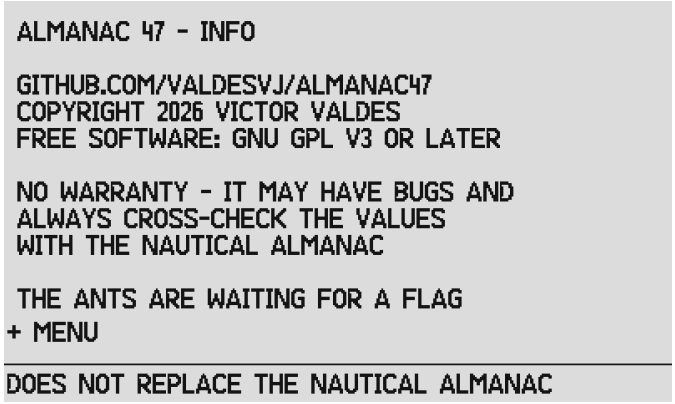
3 SKY (HORZ): full-screen chart of the 8 bodies; every 2 seconds the name of the next one is written next to its symbol (here MOON); DAY / TWILIGHT / NIGHT at the bottom.



4 ANIM (HANIM): the Sun and the Moon moving on the whole-sky chart over 12 hours (24 frames, 1 s each, 30 min apart); the first frame.



5 ALLSKY: the whole sky, over the horizon above, under the horizon below: every star, the Sun, the Moon and the planets; DAY / TWILIGHT / NIGHT.



6 INFO: the repository (github.com/valdesvj/almanac47), the licence (GNU GPL v3 or later), no warranty, and the rule: always cross-check the values with the Nautical Almanac. The warning line (DOES NOT REPLACE THE NAUTICAL ALMANAC) is on the menu and on INFO only.

9 SNAP, on the menu and on every view: a screenshot of the screen (SNAP), then the same screen again.

Below the horizon: a body with a negative Hc (in practice the Sun) has its Hc shown white on black, as the Sun in the ALMANAC example above. **X** at the top right instead of S: the date is outside the period of the matrices (FAST: 2026–2030).

6. Speed and memory

- The sky is computed once for a time and place (matrix ALMC, 73 × 4: GHA, Dec, Hc, Zn of the Sun, the Moon, planets 1–4 and stars 1–58; the sun times). Going from view to view only draws. An arrow (a new hour) computes again. ALMC and the star order SBR stay between runs.
- **v2.2.0 (Struct on Attack)** has the same calculation and the same results as v2.1.0, and the same pages with the new top bar (date, UT and DR in one line) and the hourglass. It is written with the C47 STRUCT commands and is much faster. Counted in the C47 simulator, each page with its two arrows: ALMANAC 18 %, SPLIT 42 %, SKY 53 %, ANIM 54 %, ALLSKY 40 % and the menu's arrows 45 % fewer steps than v2.1.0; and about 85 % less of the time the C47 spends looking for labels, structure steps and the place a return goes back to. Where it comes from:
- — **no GTO:** every decision and loop is a structure (IF, DO, REPEAT), the key loop of the menu one DO … ENDDO, the keys 1–6 by arithmetic on the key code;
- — **the order of the code:** the routines used most at the start of each program, the programs in the order of their use (each RTN on the C47 walks the steps from the start of its program);
- — **drawing:** the horizontal lines with PIXEL line mode, the box and the highlight in GRMOD 1 with unrolled AGRAPH, the header text kept for the run, the text routines written in place where they are called most;
- — **calculation:** Hc and Zn with four trigonometric functions, the equator of the charts a whole column at a time (M.GETM / M.PUTM), the star order as a matrix (SBR), whole numbers into text with one αIP.
- ALMA47 is 27.2 KB (8 734 steps; v2.1.0's NAVFULL 24.2 KB): the unrolled loops and the code written in place make it bigger, for speed. MOON47 is 4.5 KB. NAVINIT while it is loaded: FAST 28.5 KB, FULL 57.8 KB; the matrices FAST about 55 KB, FULL about 95 KB.
- **Accuracy, checked in the C47 firmware itself** (its PC simulator) against JPL DE421 with ERFA, 20 random instants and places each for FULL (2000–2050) and FAST (2026–2030): Sun within 0.02′, Moon 0.10′, planets 0.05′, stars 0.003′, Hc and Zn 0.08′ — the same as v2.1.0.
- History: v2.0.0 was faster and smaller than v1.1 (Horner's method, n-vectors, the nutation as matrix products, counted loops); v2.1.0 (Supercharger) the cos and sin of each series as one complex e^x, Hc and Zn from the body's direction vector. Details of every step: OPTIMIZATIONS.md.
- The C47 sends a new screen to the display only at a PAUSE, a key press or the end of the program: the SINKING box shows while a view is computed, then the complete view.
- INFO says the ants are waiting for a flag. Something lives in ALMA47 under its LBL 06 … try flag 47, then press + or an arrow.

7. Inside ALMA47: the labels

ALMA47 is 26 programs. Only the first has a name, ALMA47; each of the others starts with a numbered global label, N01, N02 … (so the names do not clash with yours), and its routines are local labels 01, 02 … The numbers are those of earlier versions: a number missing below belongs to a routine that is gone. The comments of ALMA47.p47u say what every routine does; the source is programs_struct/ in the repository (one file per program, indented).

Routines you can call yourself

Run ALMA47 with your date, time and DR, then leave it with **EXIT** (or R/S), not with 0 — 0 clears the registers. These routines read the numbers ALMA47 stores in R43–R98 when it starts (the table at the end of this section), the DR of the run (R13 longitude, R14 latitude) and the matrices of NAVINIT; they use R00–R42 for their work. JD is the C47's Julian Date (the date, then 𝔻→J) + UT / 24, a plain number.

Label	Name	Input	Output
N15	SUNA	X = JD (UT)	X GHA of the Sun, Y Dec, Z GHA of Aries
N17	SUNF	X = JD (UT)	X GHA of the Sun, Y Dec (low precision, 0.01 deg)
N25	MOO2	right after N15, same JD	X GHA of the Moon, Y Dec, Z HP ('), T SD (')
N28	PLN2	right after N15; X = planet 1-4 (Venus, Mars, Jupiter, Saturn)	X GHA, Y Dec, Z SHA, T HP (')
N29	PLN3	right after N15; X = planet 1-4	X GHA, Y Dec (a quick position, the full series only when the planet can be above the horizon)
N32	HCZ	Y = Dec, X = GHA (decimal degrees)	X Hc, Y Zn for the DR of the last run
N38	RISE	Z = JD at 0 h UT, Y = lat, X = lon (decimal degrees, N and E +)	X sunrise (UT hours; 99 = none that day)
N39	SET	Z = JD at 0 h UT, Y = lat, X = lon (decimal degrees, N and E +)	X sunset (UT hours; 99 = none)
N44	TRAN	Z = JD at 0 h UT, Y = lat, X = lon (decimal degrees, N and E +)	X meridian passage of the Sun (UT hours)
N40	NTWA	Z = JD at 0 h UT, Y = lat, X = lon (decimal degrees, N and E +)	X nautical twilight, morning (UT hours; 99 = none)
N41	NTWP	Z = JD at 0 h UT, Y = lat, X = lon (decimal degrees, N and E +)	X nautical twilight, evening (UT hours; 99 = none)
N48	SNMU	X = star number 1-58	the star's name in ALPHA

Example: ALMA47 for 23 Sep 2026, 25° 20′ N 055° 12′ E, left with EXIT. **2461307 XEQ "N15"** gives X 1.9070 (GHA of the Sun, 1° 54.4′), Y −0.1932 (Dec S 0° 11.6′), Z 182.3525 (GHA of Aries); then **XEQ "N25"** the Moon. **2461306.5 ENTER 25.333 ENTER 55.2 XEQ "N38"** gives 2.13: sunrise at 02:08 UT.

Every global label of ALMA47

Label	Name	What it does
ALMA47	NAV	graphic menu (KEY?): asks DATE UTC LAT LON, keys 1-6 a view, 9 SNAP, arrows one hour, 0 ends (CLREGS CLSTK); the only named program
N95	CALC	the sky for a new time or place: Sun, Moon, planets into ALMC; the stars as matrices SXE SXH (58 x 3), marked to compute when a page asks
N80	PTNT	TEXT: whole number in the tinyFont
N54	PDMS	TEXT: degrees and minutes 'ddd mm.m' with sign
N55	PDTS	TEXT: date 'dd-mm-yyyy' from a Julian Date
N53	PHMS	TEXT: hh:mm from hours
N51	PINS	TEXT: whole number (one ATEXT)
N79	PTTY	TEXT in the tinyFont (GRFNT 10, the charts) - Z row of the base line, Y column, X text
N50	PTXS	TEXT: ATEXT in GRFNT 21 (Didier's N03 trick) - Z row of the base line, Y column, X text
N56	PZNS	TEXT: azimuth 'ddd.d'
N52	PF1S	TEXT: number with one decimal
N57	PHLS	horizontal line (X = length in pixels)
N61	HANIM	view 4 ANIM: the Sun and the Moon moving on the whole-sky chart (24 frames)
N82	PSYS	SYMBOL of a body, 7 rows (glyphs47, AGRAPH; the charts) - Z row, Y column, X symbol

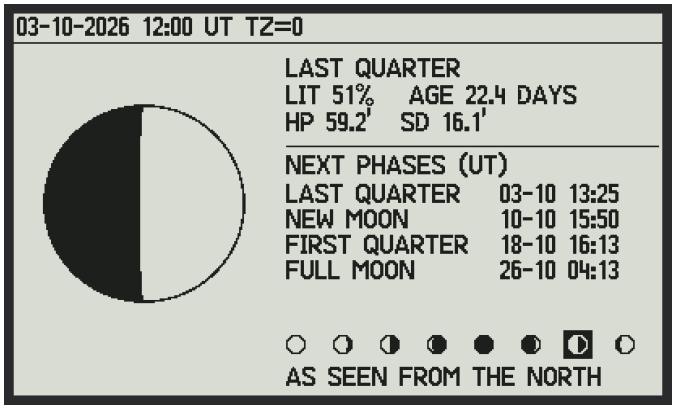
Label	Name	What it does
N62	ALLSKY	view 5 ALLSKY: whole sky, over the horizon above, under the horizon below
N29	PLN3	planet for the screens: quick position first, full series only if it can be above the horizon
N28	PLN2	planet GHA/Dec/SHA/HP: VSOP87 series with light-time, after SUNA
N32	HCZ	sight reduction for the DR (R13 lon, R14 lat): Y = Dec, X = GHA -> X Hc, Y Zn (also R09, R05), sin Hc in 'V4'
N18	SER	sum of one series matrix with the matrix functions (COS, DOT)
N19	SERT	the time vectors SV1 SV2 used by SER
N16	SUNG	Sun GHA/Dec for the sunrise iterations (tables if loaded, else SUNF)
N17	SUNF	Sun, low-precision formula (0.01 deg) for sunrise/twilight
N15	SUNA	Sun: VSOP87 series (matrices VL VB VR), nutation, aberration -> GHA, Dec, GHA Aries
N83	HDR	header of a view: date, UT, DR latitude N/S, longitude E/W (X row, Y time, Z lat, T lon; tools/navmat.py)
N69	CHCZ	Hc Zn from the cache for the body just read (else the real HCZ)
N71	CNTA	nautical twilight am from the cache (the five sun times: computed once per date and place)
N68	CSQK	star from the cache: over the horizon test from SXH (sin Hc, no trig); its angles (CSTA) the first time a page asks
N72	CRIS	sunrise from the cache
N73	CTRN	meridian passage from the cache
N74	CSET	sunset from the cache
N75	CNTP	nautical twilight pm from the cache
N77	CEQQ	celestial equator of the charts: its dots for this place (U3 SKY / SPLIT, U2 ANIM / ALLSKY), computed again only for a new place
N65	CMOO	Moon from the cache
N64	CSUN	Sun from the cache (matrix ALMC); computes the whole sky (CALC) only for a new time or place
N25	MOO2	Moon GHA, Dec, HP, SD: Meeus series (matrices ML MB MCL MCB) after SUNA
N38	RISE	sunrise time UT (SUNRISE program)
N39	SET	sunset time UT
N44	TRAN	meridian passage of the Sun, UT
N40	NTWA	nautical twilight, morning, UT
N41	NTWP	nautical twilight, evening, UT
N81	PSYB	SYMBOL of a body, 12 rows (glyphs47, AGRAPH; the tables, the Moon phases) - Z row, Y column, X symbol
N04	HORZ	view 3 SKY: horizon chart, the name of each body in turn every 2 s (+ back to the menu, arrows one hour)
N06	HALMH	view 2 SPLIT: horizon chart on top, the bodies below (8 rows)
N01	ALMF	view 1 ALMANAC: GHA, Dec, Hc, Zn table of Sun, Moon, planets, stars; twilight, rise/set, Moon
N98	CALL	ALLSKY: a star without values (-98, -99) -> all 58 at once with angles on whole columns (M.PUTM)
N48	SNMU	star name from its number
N46	PHA2	Moon phase after SUNA: illumination % and age in days (the new-moon search: one REPEAT, a second pass when the age is negative)
N63	WPLS	every view holds its screen here (KEY?): + back to the menu, up / down arrow one hour later / earlier
N84	CMN	compass row N E S W N of a chart (X row)
N90	OUT5	steps shared by several routines (tools/navmat.py outline)
N89	OUT4	steps shared by several routines (tools/navmat.py outline)
N49	TGET	almanac tables (TBL): Chebyshev lookup; used only when tables are loaded (TBL is not part of v2.2.0)
N85	CMS	compass row S W N E S of a chart, south up (X row)

The numbers in R43–R98 (stored by ALMA47 when it starts: each use is a 2-byte RCL)

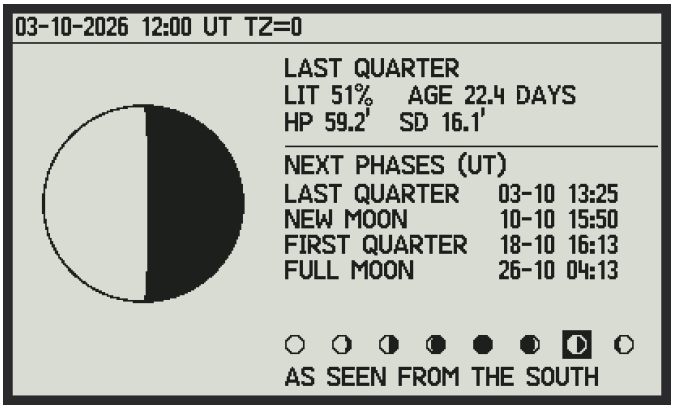
Reg.	Number	Reg.	Number	Reg.	Number	Reg.	Number
R43	1	R57	60	R71	90	R85	-1
R44	2	R58	212	R72	238	R86	184
R45	360	R59	207	R73	226	R87	82
R46	0	R60	143	R74	34	R88	5
R47	16	R61	142	R75	30	R89	280.46061837
R48	15	R62	180	R76	395	R90	2451545
R49	17	R63	176	R77	363	R91	241
R50	10	R64	100	R78	360.98564736629	R92	0.15643
R51	3	R65	27	R79	332	R93	6
R52	117	R66	4	R80	301	R94	388
R53	116	R67	51	R81	270	R95	29.530589
R54	3600	R68	26	R82	145	R96	199
R55	20	R69	25	R83	113	R97	164
R56	0.5	R70	24	R84	8	R98	150

8. MOON47: the Moon phase

MOON47 is a separate program: the Moon's phase page of the PC version on the calculator, without ALMA47 or NAVINIT. Load MOON47.p47 and **XEQ "MOON47"**. It asks nothing: the phase does not depend on where you are, so it takes the date and time from the C47's clock.



3 Oct 2026 12:00 UT, as seen from the north



the same after 0: as seen from the south

- **The page:** the Moon as a disc (the lit part filled), the phase name, % lit, age in days, HP and SD, the next new Moon, first quarter, full Moon and last quarter (date and time UT), and the eight phase symbols with today's one inverted.
- **Keys:** 0 turns the picture to the view from the south and back (the latitude only turns the picture); +/- shows the top line and the next phases in UT or in local time (LT = UT + TZ; the variable TZLT keeps the choice, as in ALMA47); any other key ends.
- **The clock is local time.** If it is not set to UT, store your offset once in the variable TZ, in hours: **4 STO "TZ"** for UT+4, **-5 STO "TZ"** for UT−5. Without TZ, MOON47 creates it with 0 and takes the clock as UT: the phase times stay right, the age and % lit are then a few hours off (% lit changes at most 0.45 % an hour). The top line shows the clock: the time UT and TZ=0, or with TZ the local time LT and your offset (TZ=+4, TZ=-5, TZ=+5:30).
- **Accuracy** against the full series of ALMA47, 2000–2050: phase times and age within 4 minutes (most within 1), HP 0.03′, SD 0.01′, % lit 0.03 %. It uses the 20 largest terms of the Moon (Meeus, ch. 47), the phases by the secant method; tests/test_moon47.py checks it.
- **v2.2.0:** one program with MOON47 its only global label (the text and symbol routines are numbered labels inside it), written with the STRUCT commands, no GTO: 4.5 KB (4.8 KB before) and about 28 % less time spent looking for labels and returns. Its work matrices M7T, M7C, M7A, M7V, M7W are set to 0 at the end (TZ stays).
- **v2.2.1:** MOON47 gives you back your stack (X … D) and LASTx when it ends (LocR 9, STOS R.00 / RCLS R.00, as ALMA47) and your stack size; it uses the registers R07–R65 and clears only those: R00–R06 and R66–R99 stay yours.
- **v2.3.0:** +/- UT or LT; the south view moved to 0.
- Free42 (MOON47.raw: your stack kept, +/- UT / LT), the old DM42, the NumWorks and the HP Prime have their own MOON47 with the same page: see their manuals.

9. NAVLITTLE: the old DM42 with the C47 firmware

The C47 firmware on the old DM42 has 64 KiB for programs, registers and matrices (the DM42n and the R47 have 256 KiB and run ALMA47). A version made only for the old DM42 keeps the Sun and the 58 stars: **NAVLITTLE** (the program ALMA47) with its own reduced **NAVINIT_LITTLE** (only the Sun series, valid 2000–2050, the nutation and the stars: 8.7 KB instead of 28.5 or 57.8 KB) and a **MOON47** of its own. No menu: the ALMANAC page straight after the inputs (the Moon and planet rows taken by the next stars), ▲ / ▼ one hour later / earlier, 9 SNAP, +/- UT or LT (as ALMA47), + ends. They are in their own zip, Almanac47_DM42_C47firmware_v2.3.0.zip, with README_DM42.txt.

Firmware	ATEXT	GRFNT	STRUCT
00.109.05.00a0.ALPHA or later: C47, R47, DM42n	in	in	in (255 a series per program)

Firmware	ATEXT	GRFNT	STRUCT
old DM42, packages 1, 2, 3	in	in	in, at most 10 IF, 10 DO and 10 REPEAT a program
old DM42, package 4 (the one in the 05.00a0.ALPHA release assets)	in	in	**out**
00.109.04.00b0 and older	out	out	out

- **Firmware:** a DM42 package with STRUCT (1, 2 or 3; see PACKAGES.txt of the firmware). With package 4 IF / DO / REPEAT only say that they are not available: keep the NAVLITTLE of v2.0.0 there.
- **Structured programming on the little memory.** The DM42 firmware keeps its tables for the STRUCT commands small: at most **10 IF, 10 DO and 10 REPEAT in each program** and 10 structures open at once (the R47, the DM42n and the C47 take 255 and 63). NAVLITTLE is written within it (its largest program: 6 IF, 2 DO, 1 REPEAT). The main MOON47 has 17 IF, so the DM42 MOON47 is a build of its own: 10 IF, 5 DO, 6 REPEAT, 7 of the IF written as a test and an XEQ of a small routine — the same steps run, the same page pixel for pixel.
- **Load NAVINIT_LITTLE alone first** (NAVINIT and its matrices together need the most memory): XEQ "NAVINIT" (MATRICES READY: SUN STARS 2000-2050), GTO "NAVINIT", DELP; then load NAVLITTLE and XEQ "ALMA47".
- **v2.2.0:** STRUCT, no GTO, its routines ordered for the searches: about 57 % less time looking for labels and returns than v2.1.0 (simulator), the same page pixel for pixel. It saves your registers R00–R35 while it runs and gives them back at the end. Memory: NAVLITTLE 7.3 KB, NAVINIT_LITTLE 8.7 KB (delete it after use), MOON47 4.5 KB; NAVLITTLE with its matrices about 31 KB.
- **v2.2.1:** NAVLITTLE and its MOON47 give you back your stack (X … D) and LASTx when they end: NAVLITTLE with LocR 45 (R.00–R.35 your registers, R.36–R.43 the stack with STOS R.36, R.44 LASTx), MOON47 with LocR 9, as on the C47. The pages are those of v2.2.0. The keys 1–8 that pick a body are in the C47 / R47 ALMA47 only.
- Checked in the simulators; not yet on an old DM42: please report the free memory and any problem.

Part 2 — the programs one by one

How each program works, with worked results

These programs give the data needed for sight reduction (GHA, SHA, declination, Hc, Zn) and for planning (sunrise, sunset, twilight, Moon phase, sky picture). They support, and do not replace, the Nautical Almanac. All results are apparent geocentric positions, as in the almanac. The time argument is UT (UT1): for a sight timed in UTC, add DUT1 (under 0.9 s).

The programs of this part are the single programs of the programs/ folder of the repository: the reference for the methods, not part of the v2.3.0 files. They are written with GTO (they also run on Free42) and draw the text with AGRAPH fonts (PTXS, PTXT, PTXB). ALMA47 of part 1 has the same calculations in a faster form (STRUCT, the v2 optimizations) and the screens of Oct 2026, with ATEXT and GRFNT; its registers and labels are its own (part 1, section 7).

Overview

Program	What it gives	Input	Output	Needs
SUNA	Sun and Aries	X = JD	X GHA☉, Y Dec☉, Z GHA♈	matrices from MATA
MATA	builds Earth-orbit and nutation matrices	—	VL, VB, VR, NU	run once
STAR	57 navigation stars + Polaris	Y = JD, X = star no.	X GHA, Y Dec, Z SHA	SUNA, matrix ST
MATST	builds star catalogue matrix	—	ST	run once
SNAM	star name	X = star no.	name on screen	—
SUNRISE	rise, set, twilight, meridian passage	Z = JD 0h, Y = lat, X = lon	X = UT hours	SUNA
PHAS	Moon phase	X = JD	X % illuminated, Y age (days)	SUNA
HORZ	picture of the sky on the horizon	Z = JD, Y = lat, X = lon	screen drawing	SUNA, STAR
ALM	Sun, Aries, Moon, planets from tables	X = t (hours in block)	X GHA, Y Dec (HP)	coefficients from the tables
TBL	almanac tables for a period (JPL), sets flag 10	—	matrices TSU…TAR	run once
TGET	Sun, Aries, Moon, planets from the tables	Y = JD, X = body 0–6	X GHA, Y Dec (Z HP, T SD)	TBL
CHZ	Hc/Zn and the inverse	T lat, Z lon, Y Dec, X GHA	X Hc, Y Zn	—
SUNSD	Sun semi-diameter	X = JD	X = SD (′)	SUNA
HALMV	sight-planning screen: chart + data	Z = JD, Y = lat, X = lon	screen	see HALMV section
ALMF	full-page almanac screen	Z = JD, Y = lat, X = lon	screen	see ALMF section
PTXS / PTXT	pixel text (big / small font)	Z = y, Y = x, X = text	drawing	—

Conventions

- **Julian Date (JD):** use the C47 date function (Date→J) and add UT/24. It must be a plain number: for example 23 Sep 2026 12:00 UT = **2461307**, and 0h UT that day = 2461306.5.
- **Signs:** latitude N +, S −; longitude E +, W −; declination negative = South.
- **Angles** come out in decimal degrees. Use →HMS or the d.mmm display to read degrees and minutes.
- **Setup order:** the v2.3.0 files do it all: load ALMA47 and NAVINIT_FULL (or NAVINIT_FAST), XEQ "NAVINIT" once, delete NAVINIT. The single programs of part 2 can still be loaded one by one (programs/ folder) for other uses.

SUNA — Sun and Aries

Computes the Sun's GHA and declination and the GHA of Aries for any instant, with no tables. It is also the engine used by STAR, SUNRISE, PHAS and HORZ.

How it works

- Time: T and τ (Julian centuries and millennia of TT since J2000; TT = UT + 69.2 s).
- Earth's heliocentric longitude L, latitude B and distance R from the VSOP87D series stored in VL, VB, VR (56 + 3 + 7 terms). The loops count rows with DSE 55.
- Sun = L + 180°, β = −B; FK5 correction.
- Nutation Δψ, Δε from the 10 largest IAU 1980 terms (matrix NU). The Moon arguments D, M, M′, F are kept in R60–R63 for PHAS.
- Apparent longitude λ = Sun + Δψ − 20.4898″/R (aberration). Obliquity ε = 84381.448″ − 46.815″·T + Δε.
- Dec = asin(sin β cos ε + cos β sin ε sin λ); RA from →POL.
- GHA♈ = GMST + Δψ·cos ε; GHA☉ = GHA♈ − RA (mod 360).

Inputs and outputs

Input	X = JD (UT1)
Output	X = GHA☉, Y = Dec☉, Z = GHA♈ (degrees). Also R73 = Sun distance R (au): SD☉ = 15.99′ ÷ R73.
Registers	R50–R55, R60–R67, R70–R81

Example with results

23 Sep 2026, 12:00:00 UT: key **2461307 XEQ SUNA**

		Result	Deg / min	Almanac page
X	GHA☉	1.9069849	1° 54.4′	1° 54.4′
Y	Dec☉	-0.1931700	S 0° 11.6′	S 0° 11.6′
Z	GHA♈	182.3525029	182° 21.2′	182° 21.2′
R73 →	SD☉	1.003445 au	15.94′	15.94′

Notes

- Accuracy vs JPL/IAU reference 2000–2050: GHA ≤ 0.016′, Dec ≤ 0.007′, GHA♈ ≤ 0.003′. Verified on the C47.
- Coefficients never need updating. ΔT (69.2 s, line 4) affects the Sun by only 0.0007′ per second.

MATA and MATST — setup programs

They contain no calculation. Each creates matrices and fills them with coefficients, so you don't have to type about 590 numbers by hand. Run each once; afterwards the programs can be deleted.

Program	Creates	Contents
MATA	VL 56×4, VB 3×4, VR 7×4	VSOP87D Earth series rows: k, A, B, C (term = A·cos(B + C·τ)·τ^k)
MATA	NU 10×9	IAU 1980 nutation: multipliers of D, M, M′, F, Ω and coefficients S, S′, C, C′
MATST	ST 58×4	Star J2000 RA, Dec (deg) and proper motions (mas/yr), stars 1–57 in almanac order, 58 = Polaris

Memory kept afterwards: about 5.7 KB (VL, VB, VR, NU) + 3.7 KB (ST).

STAR — navigation stars

Gives GHA, SHA and declination of any of the 57 almanac navigation stars or Polaris (58).

How it works

- Runs SUNA for the instant (nutation, obliquity, Sun longitude, GHA♈). Entry STR2 skips this when SUNA has already run (used by HORZ).
- Reads the star's J2000 position and proper motion from ST and applies proper motion.
- Precession J2000 → date with the IAU 2006 angles ζ, z, θ.

- Converts to ecliptic coordinates and adds nutation Δψ and annual aberration there (accurate even for Polaris), then back with the true obliquity.
- SHA = 360° − RA; GHA = GHA♈ + SHA.

Inputs and outputs

Input	Y = JD (UT1), X = star number 1–58
Output	X = GHA, Y = Dec, Z = SHA
Registers	R56–R59, R68–R69, R82–R89 (plus SUNA's)

Example with results

23 Sep 2026, 12:00 UT: key **2461307 ENTER 18 XEQ STAR** (Sirius), and 58 for Polaris

Star	GHA	SHA	Dec
18 Sirius	80° 46.2′	258° 25.1′	S 16° 44.9′
58 Polaris	135° 19.7′	312° 58.5′	N 89° 22.4′

Notes

- Accuracy 2025–2050: 57 stars ≤ 0.003′; Polaris ≤ 0.04′ in hour angle (tiny on the sky). Checked against USNO for 30 stars: ≤ 0.009′.
- Star numbers are the official almanac numbers; the list in STAR_list.txt gives them by number and alphabetically.

SNAM — star names

Shows the name of a star from its number: key **18 XEQ SNAM** → display "18 Sirius". It stores the number in R49, jumps with GTO IND 49 to local label 01–58, loads the name as text and shows it with AVIEW. The stack is unchanged.

To show the name automatically after STAR, add RCL 82 · XEQ "SNAM" · R↓ before STAR's last RTN.

SUNRISE — rise, set, twilight, meridian passage

Times of sunrise, sunset, civil and nautical twilight and meridian passage for any position, directly in UT.

How it works

- Starts from a first guess (06h for morning events, 18h for evening, 12h for transit).
- Computes the Sun at that time with SUNF, a low-precision formula (Astronomical Almanac, within 0.01°; the tables when loaded), then the hour angle at which the Sun reaches the event altitude: cos H = (sin h_0 − sin φ sin δ)/(cos φ cos δ).
- Corrects the time by (target LHA − actual LHA)/15 and repeats until the correction is under about 1 s (3–4 passes). With SUNF the times are within about 15 s of the full SUNA, well inside the minute shown.
- Event altitudes h_0: rise/set −0°50′ (upper limb, refraction), civil −6°, nautical −12°. If the Sun never reaches it, the result is 99.

Inputs and outputs

Input	Z = JD at 0h UT of the day, Y = latitude (N+), X = longitude (E+, W−)
Labels	NTWA / CTWA (morning twilight), RISE, TRAN, SET, CTWP / NTWP (evening twilight)
Output	X = UT in decimal hours (→HMS to read); 99 = no event
Registers	R90–R99

Example with results

23 Sep 2026, latitude 10° N, longitude 0°: key **2461306.5 ENTER 10 ENTER 0**, then the label

Label	Event	UT	Almanac page (10° N)
NTWA	nautical twilight am	05:03:47	05 04
CTWA	civil twilight am	05:28:09	05 28
RISE	sunrise	05:49:09	05 49
TRAN	meridian passage	11:52:22	11 52 22
SET	sunset	17:55:28	17 56

Label	Event	UT	Almanac page (10° N)
CTWP	civil twilight pm	18:16:27	18 17
NTWP	nautical twilight pm	18:40:49	18 41

Medellín (6.25° N, 75.57° W): sunrise 10:51:24 UT (05:51 local), sunset 22:57:41 UT (17:57 local). 89.5° N: RISE gives 99 (no sunrise).

Notes

- Checked against an exact search on the JPL/IAU reference: agrees to the second.
- The almanac holds the noon declination fixed for the whole day, so its times can differ by up to a minute (sunset and civil pm above). The program is the more exact one.

PHAS — Moon phase

Percentage of the Moon's disk illuminated and the age of the Moon (days since new moon), as printed in the almanac.

How it works

- Runs SUNA, which leaves the mean Moon arguments D, M, M′, F in R60–R63.
- Phase angle i (Meeus 48.4) → illuminated fraction k = (1 + cos i)/2.
- Age: days since the mean new moon, corrected to the true new moon with Meeus ch. 49 terms; near new moon it checks the neighbouring lunation.

Inputs and outputs

Input	X = JD (UT1)
Output	X = % illuminated, Y = age (days)
Registers	R41–R48

Example with results

23 Sep 2026, 12:00 UT: key **2461307 XEQ PHAS** → X = **89.1 %** (waxing gibbous), Y = **12.36 days**. 10 Oct 2026 12h: 0.0 %, 29.36 d (new moon that evening); 11 Oct 12h: 0.8 %, 0.84 d.

Notes

- Accuracy vs JPL: illumination ≤ 0.3 %, age ≤ 2 minutes.

HORZ — sky picture on the horizon

Draws a chart of the whole sky: azimuth Zn across (N–E–S–W–N, ticks every 30°), altitude Hc up (ticks and labels at 30°, 60°, 90°), the celestial equator dotted, and the same bodies as ALMF, HALMV and ALMT: the Sun (☉), the Moon and planets above the horizon, and the brightest navigation stars higher than 10° until 10 bodies (★ with its number). Then an info line for each body (name, ZN, HC), 3 s each, without end: after the last the first again. UT is at the top right, the DR under it; + returns to the NAV menu, the arrows move one hour. The Sun below the horizon is not drawn; its info line shows HC with a minus sign.

How it works

- Runs SUNA once, then STAR through entry STR2 for stars 1–58 (fast: no repeated SUNA). The celestial equator is drawn as dots every 2° of hour angle.
- For each object: LHA = GHA + λ; Hc = asin(sin φ sin δ + cos φ cos δ cos LHA); Zn from →POL of (−cos δ sin LHA, cos φ sin δ − sin φ cos δ cos LHA).
- Screen: column = 20 + Zn × 375/360; row = 225 − Hc × 200/90 (from the top; converted for the C47, which counts rows from the bottom). Only objects with Hc > 0 are drawn.
- Text (N E S W, 30 60 90, star numbers, the info line) is drawn with PTXT, the small AGRAPH font (PTNS for integers, PT1 for one decimal); the symbols with PTXS (* and @). Lines and dots use PHLS and PIXEL.

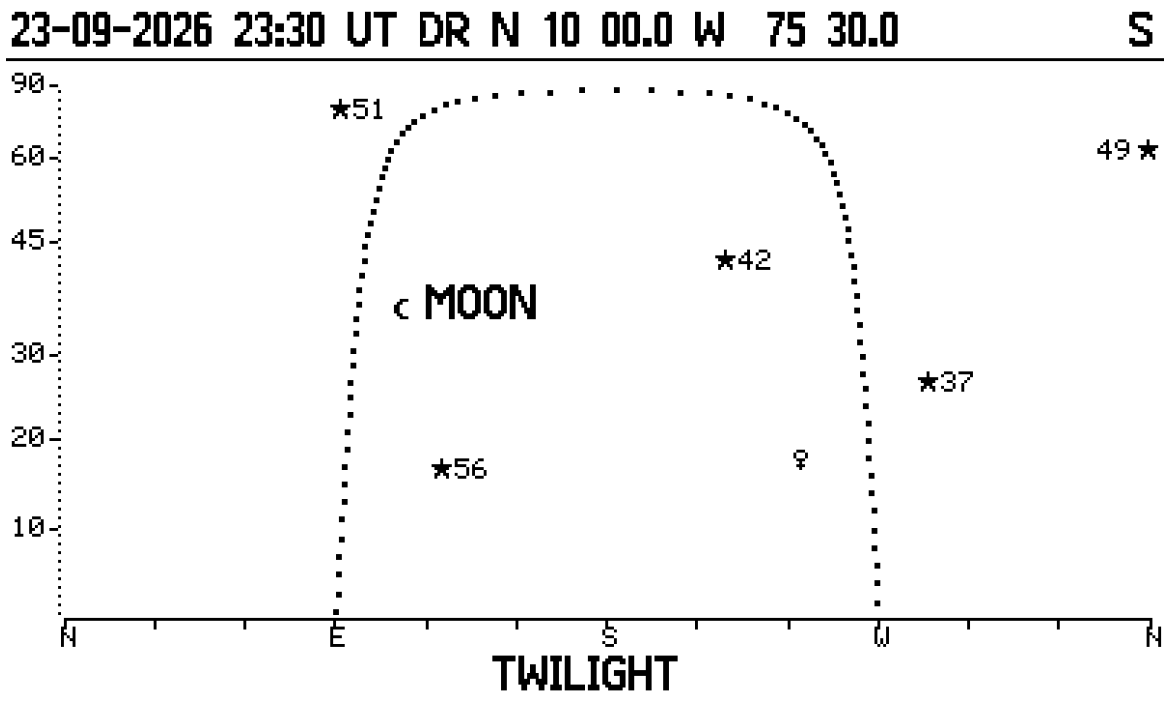
Inputs and outputs

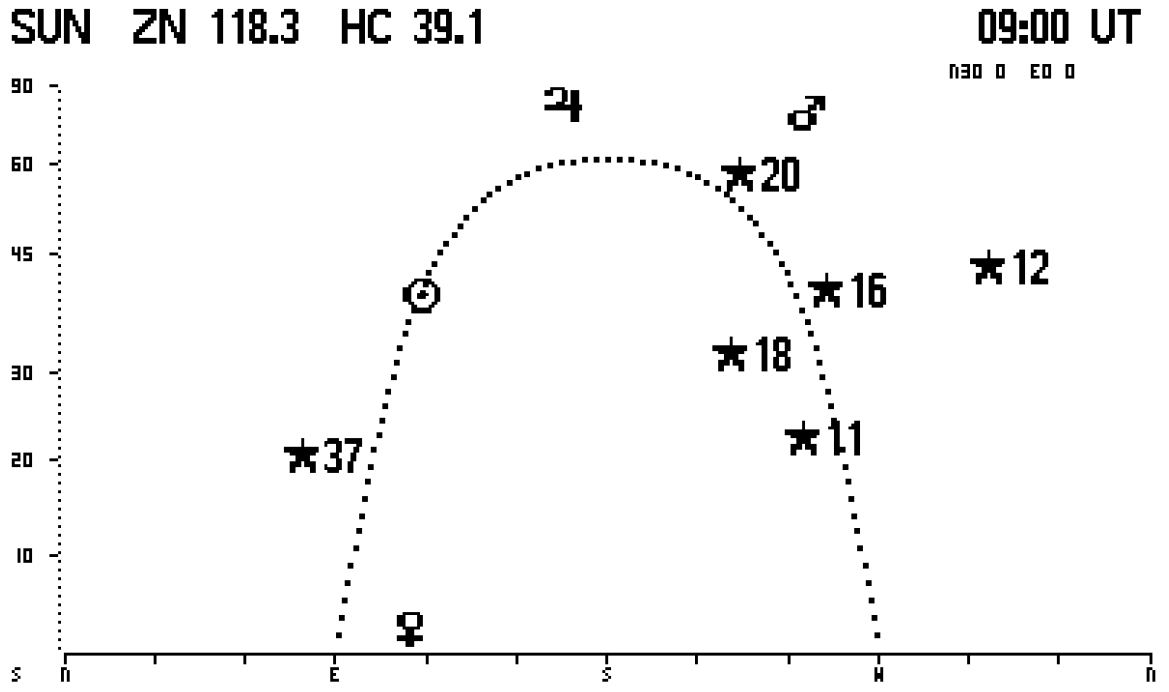
Input	Z = JD (UT1), Y = latitude (N+), X = longitude (E+)
Output	drawing on the 400×240 screen; R96 = Hc and R97 = Zn of the last object
Needs	SUNA, STAR (STR2), CHZ, PTXT, PTXS

Input	Z = JD (UT1), Y = latitude (N+), X = longitude (E+)
Registers	R90–R99, R36, R37, R42, R44, R86, R93

Example with results

23 Sep 2026, 30° N 0° E. Night 03:00 UT: key **2461306.625 ENTER 30 ENTER 0 XEQ HORZ**; day 09:00 UT: 2461306.875. Simulated screens:





Notes

- Hc and Zn agree with the reference within 0.0013′; Sun Hc/Zn match USNO (88.0833° / 264.215° vs 88.0834° / 264.214°).
- Stars are drawn even in daylight; they are only observable in twilight.

HALMV — sight-planning screen (chart + almanac data)

One screen for twilight star sights: on the left a picture of the sky (Hc up, Zn across) with the Sun, Moon, planets and the chosen stars; on the right the date and time, your DR, GHA Aries, the Sun's GHA, Dec and SD, a table of Hc and Zn (Sun, then the Moon and planets that are above the horizon, then the brightest stars, 10 rows), the day's sun times, and the Moon's phase, HP and SD.

How it works

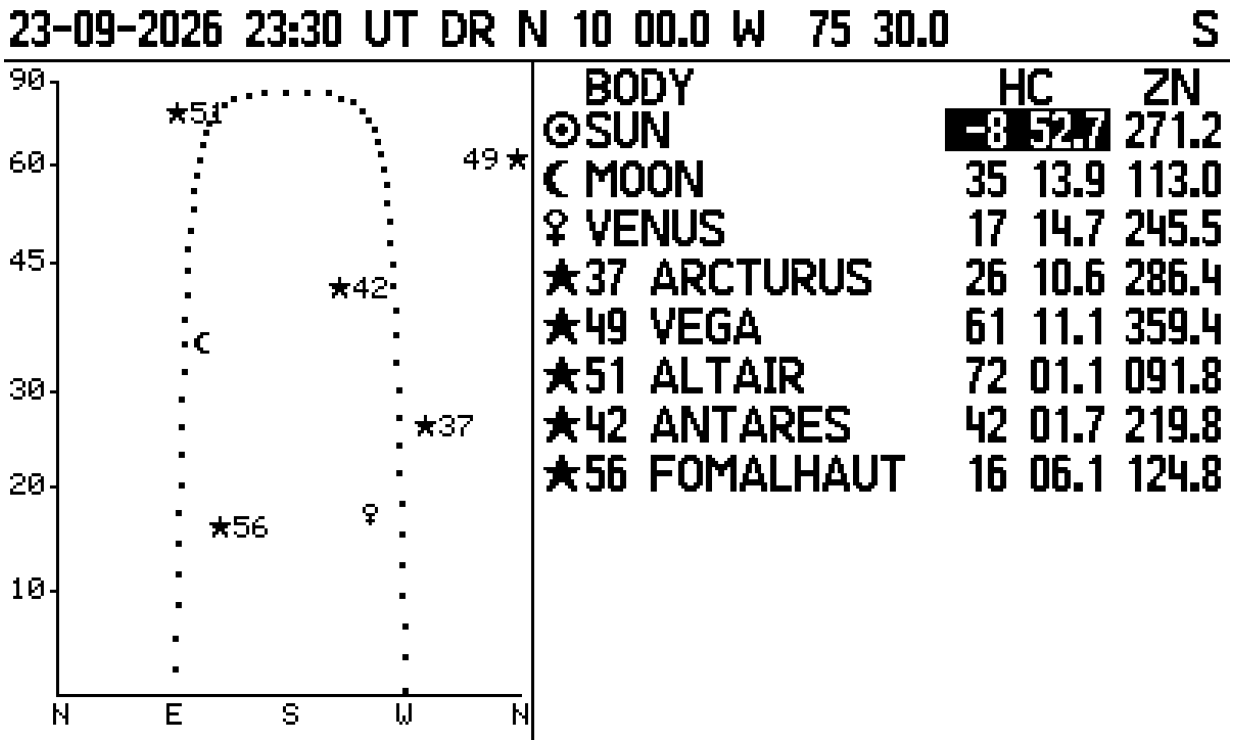
- First computes the sun times with SUNRISE (NTWA, RISE, TRAN, SET, NTWP for the UT date of the JD) and the Moon with PHAS, and keeps them in R13–R19.
- Runs SUNA once (Sun GHA, Dec, GHA Aries; SD = 15.994′ / R from R73), then STAR through entry STR2 for stars 1–58.
- Moon (MOO2) and planets (PLN3) are computed without repeating SUNA. PLN3 first places each planet from its mean orbit (within 0.2°); if it is more than 1° below the horizon the full series is skipped, otherwise the series run once with the light time from that first distance (same result as PLAN within 0.0002′). Symbols: ☾ Moon, ♀ Venus, ♂ Mars, ♃ Jupiter, ♄ Saturn. Stars: in order of brightness (SBRT), only those higher than 10°, until the table has 10 rows.
- Hc and Zn: CHZ subroutine HCZ. Chart: column = 18 + Zn × 178/360, row = 14 + Hc × 200/90; the celestial equator is dotted every 3° of hour angle. South latitude: chart centred on N (S W N E S).
- All text and numbers are drawn with AGRAPH through PTXS (the C47 status-bar font, 12 px) and PTXT (small 3×5): strings, PINS, PDMS (deg + min), PZNS (bearing), PHMS (hh:mm), PF1S and PDTS (date from JD). Star names from SNMU. The screen is held by WPLS (+ back to the menu, arrows one hour).

Inputs and outputs

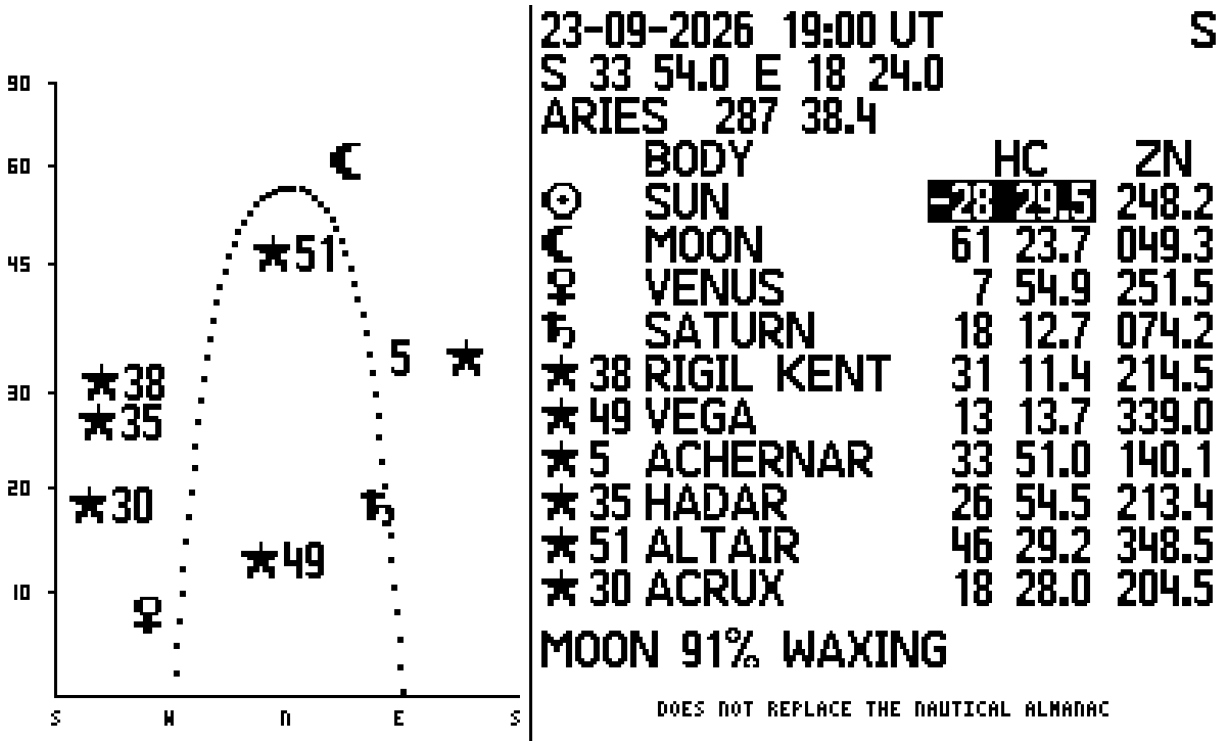
Input	Z = JD (UT1), Y = latitude (N+), X = longitude (E+)
Screen	left: chart; right: DR, date + UT, Aries, Sun SD, Sun GHA/Dec, table BODY / HC / ZN, naut. twilight AM/PM, rise/set, mer pass (UT), Moon % + WAXING/WANING/FULL/NEW + age
Needs	SUNA, STAR, CHZ, SUNRISE, PHAS, MOON, PLAN, SBRT, SNMU, PTXS + all matrices (MATA, MATST, MATM, MATP)
Registers	R10–R29, R37, R40–R48 (+ those of the programs it calls)

Example with results

23 Sep 2026, 23:30 UT, DR 10° 00′ N 075° 30′ W: key **2461307.4792 ENTER 10 ENTER −75.5 XEQ HALMV**. Simulated screen:



Southern example, 33° 54′ S 018° 24′ E, 19:00 UT (chart centred on N):



Notes

- Sun times checked against the JPL-based reference for this position: naut. twilight 10:05:46 / 23:42:41, rise 10:51:08, set 22:57:20, mer pass 16:54:18 UT — the screen rounds to the minute as the almanac does.
- All times are UT for the UT date of the JD, at the DR entered; sea level, standard refraction (as the almanac). West of Greenwich an evening event after 24h UT shows as 00:xx (next UT day).
- Hc and Zn are computed from the DR: use them to identify stars and pre-set the sextant. Near the chart edges two star numbers may overlap; the table is always exact.
- Sun SD = 959.63″/R; it can differ by 0.1′ from the printed page, which gives one SD per three days.

ALMF — full-page almanac screen

The same data as HALMV without the chart, laid out like an almanac page: GHA and Dec as well as Hc and Zn, with the same 10-row rule as HALMV (Sun, then the Moon and planets above the horizon, then the brightest stars higher than 10°), the sun times, Moon phase, Moon HP and SD, and Sun SD. The last line reads DOES NOT REPLACE THE NAUTICAL ALMANAC (also on HALMV).

How it works

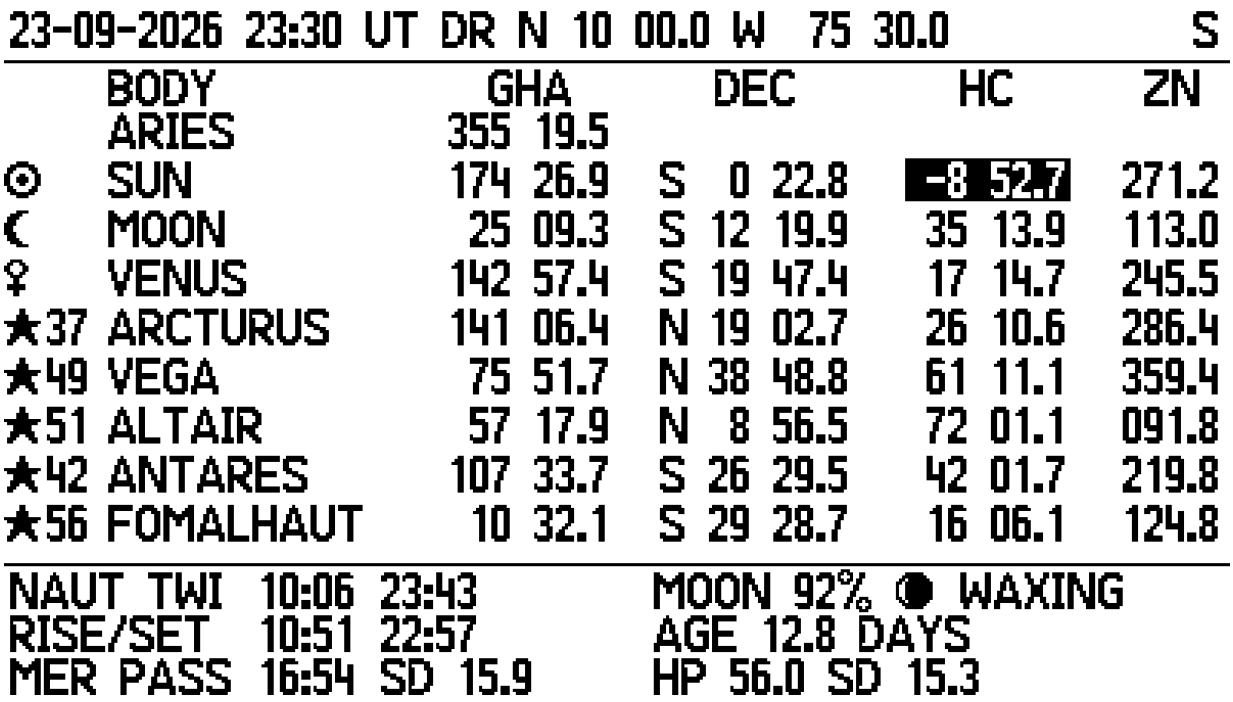
- Same calculations and table rule as HALMV.
- Top line: date, UT, DR and GHA Aries. Table: symbol, number, name, GHA, Dec, Hc, Zn. Bottom left: nautical twilight and rise/set in AM/PM columns and meridian passage; bottom right: Moon % with WAXING, WANING, FULL (shown 100 %) or NEW (0 %), age, Sun SD.

Inputs and outputs

Input	Z = JD (UT1), Y = latitude (N+), X = longitude (E+)
Needs	same programs as HALMV
Registers	same as HALMV

Example with results

Same instant and DR as the HALMV example: key **2461307.4792 ENTER 10 ENTER −75.5 XEQ ALMF**. Simulated screen:



Notes

- GHA and Dec are the almanac values for that instant (Sun ≤ 0.02′, stars ≤ 0.003′ against JPL); read them directly for sight reduction.
- HALM is a third layout of the same data: chart on the top half, table (GHA, Dec, Hc, Zn) on the bottom half.

Supporting programs for the screens

Program	Use	Stack
PTXS	the C47 status-bar font (12 px); * = star, @ = sun, also % ; entries PINS, PDMS, PZNS, PHMS, PF1S, PDTS	Z = y, Y = x, X = text or number → Y = y, X = next x
PTXT	small 3×5 pixel font, same characters and stack	as PTXS
SNMU	star name in capitals, returned in X (no display)	X = star no. → X = name
SUNSD	Sun semi-diameter 15.994′ / R	X = JD → X = SD (′)
CHZ / HCZ / DHA	Hc and Zn; inverse gives Dec, LHA, GHA	CHZ: T lat, Z lon, Y Dec, X GHA → X Hc, Y Zn

ALM — table method (Sun, Aries, Moon, planets)

Evaluates short Chebyshev polynomials whose coefficients you key in from the tables in C47_GHA_Dec_Almanac.pdf or the Excel file. It is the program for the Moon and the planets (Venus, Mars, Jupiter, Saturn).

How it works

- Each table line covers a block: 8 days (192 h) for Sun, Aries and planets; 1 day (24 h) for the Moon.
- x = 2t/span − 1, where t = hours since the start of the block.
- CHEB sums the series with the Clenshaw recurrence; ALM does it for GHA (R10…) and Dec (R20…); HP does the Moon's horizontal parallax (R30–R33).

Inputs and outputs

Store	R01 = n (5, Moon 6), R02 = span (192, Moon 24), GHA coefficients R10…, Dec R20…, Moon HP R30–R33
Input	X = t (hours since the block start, UT1)
Output	X = GHA, Y = Dec; XEQ HP → horizontal parallax (arcmin)

Store	R01 = n (5, Moon 6), R02 = span (192, Moon 24), GHA coefficients R10…, Dec R20…, Moon HP R30–R33
Registers	R01–R09, R10–R33, R40–R41

Example with results

Sun, 23 Sep 2026 12:00 UT: block 2026-09-17, t = 6 × 24 + 12 = 156

	c0	c1	c2	c3	c4
GHA	1621.68475	1440.35480	-0.00108	-0.00024	-0.00000
Dec	0.77751	-1.55320	-0.00149	0.00024	-0.00000

Result: GHA 1° 54.4′, Dec S 0° 11.6′ (same as SUNA).

Moon, 23 Sep 2026 19:00 UT: block 2026-09-23, t = 19

	c0	c1	c2	c3	c4	c5
GHA	217.86332	174.54982	0.01758	-0.00081	-0.00002	0.00000
Dec	-14.82363	2.56120	0.04565	-0.00100	-0.00001	-0.00000
HP	55.738	0.303	0.004	-0.000		

Result: GHA 319° 40.8′, Dec S 13° 20.6′, HP 55.91′ (SD ≈ 0.2725 × HP = 15.24′).

Notes

- Accuracy 0.002′ (HP 0.001′). Tables valid 1 Sep 2026 – 31 Dec 2027; later years with c47_almanac_generator.py.
- Not yet run on the C47: check RCL IND and DSE with the Sun example.

TBL and TGET — almanac tables, and the T / S switch

Precise GHA and Dec for a limited period, taken from Chebyshev tables fitted to the JPL ephemeris (the same source as the printed almanac). Once the tables are loaded, the Sun, Aries, Moon and planets on every screen come from them automatically, the screens need about half the program steps, and each screen shows **T** (tables) or **S** (series). Outside the table period everything falls back to the series (SUNA, MOON, PLAN), which work for any date.

How it works

- **Making the tables (PC):** tools/almanac/c47_almanac_generator.py computes apparent GHA and Dec from JPL DE421 with the IAU 2006/2000A precession-nutation (ERFA) and fits a Chebyshev polynomial to each block: 8 days (5 terms) for the Sun, Aries and planets, 1 day (6 terms, plus 4 for HP) for the Moon. The tables for 1 Sep 2026 – 31 Dec 2027 are in the Excel file.
- **Turning them into a C47 program:** tools/almanac/tab2c47.py START END writes TBL.txt (convert with rejig). TBL builds one matrix per body — TSU Sun, TVE Venus, TMA Mars, TJU Jupiter, TSA Saturn, TMO Moon, TAR Aries — and sets flag 10. Row 1 of each matrix holds the JD of the first block, the block length in days, the number of blocks and the number of terms; each following row holds the coefficients of one block.
- **TGET** finds the block for the JD, computes x = 2 (JD − block start) / length − 1 and sums the series (Clenshaw) for GHA (MOD 360), Dec and, for the Moon, HP; SD = 358473400 sin HP / 6378.14 / 60.
- **The switch (flag 10 set):** SUNA replaces the Sun GHA/Dec (R81, R77) and GHA Aries (R80) with the table values; the stars keep their own calculation with that GHA Aries. SUNG, used by the SUNRISE iterations, takes the Sun from the tables (outside them, the low-precision SUNF). MOO2 and PLN2 call TGET first; if the date is outside the table they compute the series. MOO2 sets flag 11 when the Moon came from the tables.
- **The letter on the screens:** T = flag 11 (the Moon, and so everything except the stars, from the tables), S = series. ALMF and HALMV bottom right, HORZ and HORZS bottom left, ALMT at the end of the first line. On days at the edge of the period (the Moon table starts or ends one day before the 8-day blocks) the screen shows S while the Sun and planets may already come from the tables.

Inputs and outputs

TBL	run once: XEQ TBL → builds the matrices, sets flag 10, shows “TBL 26-09-2026 TO 31-01-2027”. The TBL program can then be deleted; the matrices stay.
TGET in	Y = JD (UT1), X = body: 0 Sun, 1 Venus, 2 Mars, 3 Jupiter, 4 Saturn, 5 Moon, 6 Aries
TGET out	X = GHA, Y = Dec (degrees); Moon also Z = HP, T = SD (arcmin); Aries Y = 0. X = −1 when the date is outside the table.
Flags	10 = tables loaded (CF 10 = use the series only); 11 = Moon from the tables (T on the screens)
Registers	TGET: R00–R09, R35–R39 (the MOON/PLAN work registers)
Programs	TBL, TGET and the updated SUNA, SUNRISE, MOON, PLAN, ALMF, HALMV, HORZ, HORZS, ALMT

Example with results

Cape Town (33° 54′ S, 18° 24′ E), 23 Nov 2026 09:00 UT, JD 2461367.875. Tables (TGET) against the series programs:

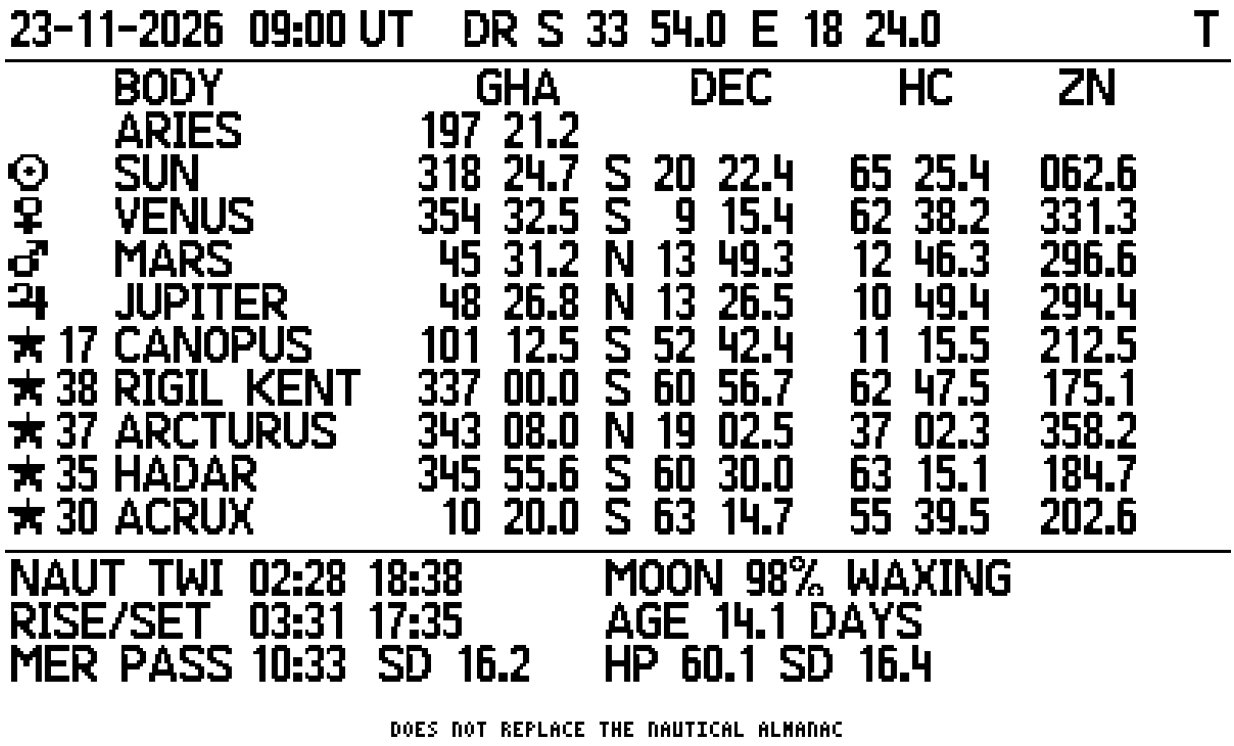
Body	GHA tables	Dec tables	GHA series	Dec series	ΔGHA	ΔDec
Sun	318° 24.7′	S 20° 22.4′	318° 24.7′	S 20° 22.4′	0.002′	0.004′
Moon	157° 34.8′	N 20° 40.0′	157° 34.8′	N 20° 40.0′	0.006′	0.016′
Venus	354° 32.5′	S 9° 15.4′	354° 32.4′	S 9° 15.4′	0.023′	0.018′

Moon HP 60.10′ and SD 16.38′ from the tables (series 60.10′, 16.38′). GHA Aries 197° 21.2′. Outside the table (1 Mar 2027) TGET returns −1 for the Moon.

Program steps for the same screens, series → tables:

Screen	Series	Tables	Saving
ALMF	100,142	38,490	62 %
HALMV	109,732	48,080	56 %
HORZS	67,201	27,990	58 %

ALMF with the tables — note the T in the bottom right corner:



Notes

- Differences between tables and series are the series' own error: at most about 0.005′ for the Sun, 0.02′ for the planets and 0.07′ for the Moon. On the screens this changes at most the last digit (0.1′).
- One TGET call takes about 265 steps (Moon 400) against 2 000–12 000 for the series; that is where the time saving comes from.
- The tables expire: load the next period in time (a quarter or a year per file, depending on memory). TBL for 26 Sep 2026 – 31 Jan 2027 holds 3 054 numbers, about the size of MATP.
- ΔT (TT − UT1) is fixed in the generator (69.2 s). One second of error moves the Moon by about 0.01′.

- The PC version (python/native/c47pc.py) reads the same tables (by default build/TBL_5.txt, 1 Oct 2026 – 30 Sep 2031) and gives the same screens (box “Almanac tables”, or --tables / --series).

Using the results for Hc and Zn

- GHA and Dec at the sight time (UT = UTC + DUT1): SUNA (Sun), STAR (stars), ALM (Moon, planets).
- LHA = GHA + λ (E +), mod 360.
- Hc = asin(sin φ sin δ + cos φ cos δ cos LHA).
- Zn = atan2(−cos δ sin LHA, cos φ sin δ − sin φ cos δ cos LHA), mod 360.
- Intercept = Ho − Hc, toward Zn if positive. The C47 key →HcZ may do these steps directly.
- For Ho you still need index error, dip, refraction, SD (☉ from R73, ☾ from ALM HP) and the almanac's planet corrections.

Python and NumWorks version

nav.py + navdata.py contain the same methods (SUNA, STAR, SUNRISE, PHAS) for a computer or a NumWorks. The menu asks for year, month, day and UT (hh:mm:ss) and computes the JD itself. Results are identical to the C47 programs (for example Sun 23 Sep 2026 12h: GHA 1° 54.4′, Dec S 0° 11.6′).

PC version (c47pc 1.5)

python/native/c47pc.py shows the same screens in a window on the PC (GTK 3), with its own calculations and no calculator programs, identical pixel for pixel to the C47. − / + step the UT (keys or the buttons next to Now UTC) by 1 s to 999 days (1 h by default); hold the key to watch the sky move. It also writes a screen to PNG (--png) or the ALMT lines as text.

Register map

Program	Registers
SUNA	50–55, 60–67, 70–81
STAR	56–59, 68–69, 82–89
SUNRISE, HORZ	90–99
PHAS	41–48
SNAM	49
ALM	01–09, 10–33, 40–41
CHZ	91–92, 94–97
PTXS / PTXT	30–36
HALMV, ALMF, HALM	10–29, 37, 40–48 (work registers, results of SUNRISE/PHAS kept in 13–19)

C47 notes

- The C47 does not treat →DEG and flag 77 like the HP-42S; the programs avoid both (verified on the C47 with SUNA).
- Memory: ALMA47 8 734 steps (27.2 KB); the FULL matrices about 6 000 numbers (FAST about 3 400), plus the cache matrix ALMC (73 × 4) and the star order SBR (ALMA47 deletes the ALMQ that NAVINIT makes).
- Keep the printed Nautical Almanac as the primary reference; use the calculator as support and cross-check.